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Lectures 5 & 6: Randomized Trials

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Where are we and why start here?

EPI Studies

What is the aim of the study?

Descriptive
- Cross-sectional (Survey)

Analytic
- Experimental
  - RCTs
    - Individual-level
    - Group-level (cluster)
    - Cross-over designs
    - Multi-arm trials vs. Factorial designs
  - Quasi-Experimental Designs
  - **Exposure assigned**
- Observational analytic
  - Cohort study
  - Cross-sectional (Analytic)
  - Case-control study
  - **Exposure not assigned**

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Lecture Objectives

To discuss the following features of randomized trials:

1. Overview of trial characteristics
2. Analytic approaches
3. Problems and limitations in trials
4. Biases

5. Different types of Trials!

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Review: Overview of RCTs

- Ideal study design for evaluating the efficacy (and the side effects!) of a new form of treatment/intervention
- Begins with a defined population that has been randomized to receive either current treatment(standard of care)/placebo or new treatment
- Participants are followed over time and monitored for primary and secondary outcomes
- Selection of participants and randomization into treatment groups is critical for study validity and *determination* of potential generalizability
- Discuss:
  - Advantages of RCTs?
  - Disadvantages of RCTs?

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When to Use RCTs

- Required by government regulatory bodies as the basis for approval decisions for new treatments/interventions
- Do researchers have a specific, targeted study question about a potential treatment or intervention?
  - RCTs are the "Gold Standard" for *attempting* to show causation
- Do the conditions exist such that participants can be appropriately recruited, enrolled, randomized, and followed for the duration of the study?
- Do the conditions exist such that data can be gathered, cleaned, analyzed, and reported accurately

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Key characteristics of "controlled" trials

- **Eligibility**: strict inclusion/exclusion criteria
- **Randomization:**
  - minimizes confounding by known and unknown factors if sample size is large
  - comparability of randomized sample
- **Blinding/masking:**
  - comparability of collected Exposure and Disease information
  - minimizes differential misclassification of exposure and effect data
- **Control:**
  - placebo or standard of care
  - comparability of trial circumstances/conditions
  - minimizes misspecification of the effect measure
- **Endpoints:** primary vs. secondary
- **Adverse events** and **stopping rules**
- **Analyze the data!**

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Randomized controlled trial example: Coffee and Pancreatic Cancer

- Based on people diagnosed with pancreatic cancer between 2015 and 2021, although diagnosis and treatment of pancreatic cancer is improving, the five-year survival rate ranges for 44% to 3% based on stage at time of diagnosis (https://www.cancer.org/cancer/pancreatic-cancer/detection-diagnosis-staging/survival-rates.html)
  - Thus, detecting modifiable risk factors that may serve as primary prevention for this cancer necessary
  - Coffee is a popular beverage globally
  - **Objective:** *To study the association between coffee consumption and pancreatic cancer risk*

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Randomized controlled trial example: Identifying the study question

What makes a good question?

- Specific
- Applicable to and important for the target population
  - not just study sample.
  - If we answer this question well, will it lead to improved health in the target population? How? How quickly?

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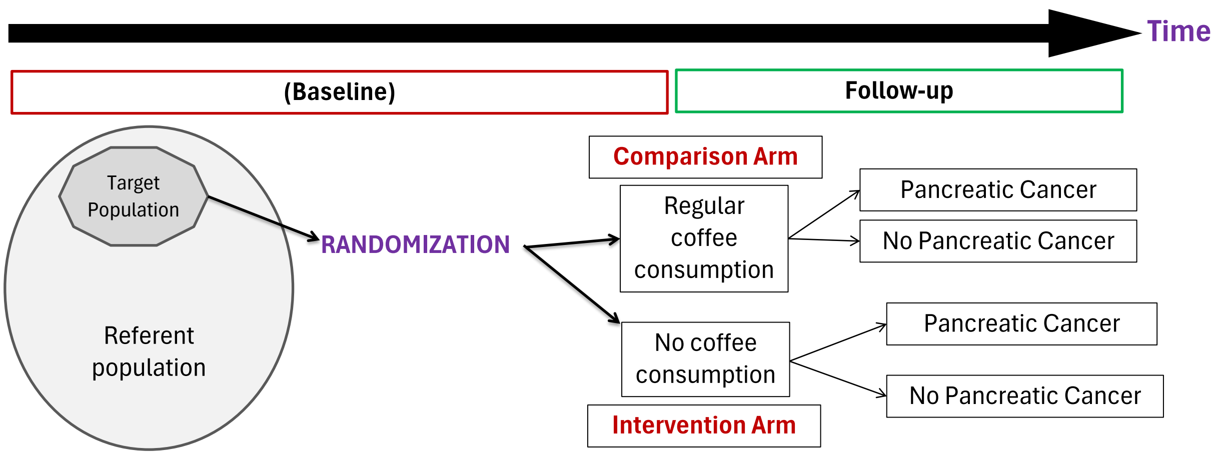
Randomized controlled trial example: Coffee and Pancreatic Cancer

RQ: Does eliminating coffee consumption reduce risk for pancreatic cancer development?
Hypothesis: Eliminating coffee consumption decreases pancreatic cancer *incidence*



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Overall steps in analysis of RCT data

- Complete all data quality checks
- Freeze the data set – no more changes
- **Conduct analysis**, often with a statistician blinded to group identities
  - *All analyses are pre-specified*
- And then the big reveal: Unblind the investigators!

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Let's first consider some RCT outcome data scenarios

- Overview: Test new drug (A) against the standard treatment (B)
  - Randomize 1000 people (500 to each treatment arm); Follow-up period = 1 year and assume no LTFU in either group!
    - **Scenario 1:**
      - End of f/u: Group A=10 outcome events and Group B=5 outcome events
      - What is the risk in group A? The risk in group B?
    - **Scenario 2:**
      - End of f/u: Group A=10 outcome events; 10 discontinue drug use / Group B=5 outcome events; 40 discontinue drug use
    - **Scenario 3:**
      - End of f/u: Group A=10 outcomes events, 10 discontinue drug use, 10 shift to drug B.
      - End of f/u: Group B=5 outcome events, 30 discontinue drug use, 8 shift to drug A.

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Main approaches for analyzing trial data

- Two main approaches:
  - **Intent to treat (ITT) analyses:** *Comparisons are made between the outcomes for the complete groups assigned to different treatment regimens, irrespective of the extent of departure from the treatment schedules laid down in the protocol*
    - analyzing by <u>assignment</u>, so therefore exploring the <u>effect of assignment</u> and not actual receipt of the drug
  - **Compliance-corrected analyses:** *Comparisons are made between the outcomes based on compliance with intervention (i.e.* analyze people by *what they did*)
    - AKA: "as-treated," or "adherence-corrected"
    - exploring the effect of actual receipt of the drug, not of assignment
- Why do these distinctions matter?

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These distinctions matter because..

- RCTs are a 'gold standard' for assessing causal inference, if they can demonstrate 4 arguments:
  - **Temporality**
  - **Consistency of treatment assignment**: In a placebo-controlled RCT, participants are assigned to treatment or placebo in the same way, so there is no variation in treatment assignment
  - **Positivity of treatment assignment:** All RCT participants have a nonzero probability of being assigned to each treatment and placebo AND getting that treatment.
  - **Exchangeability in treatment assignment:** For every female participant between 45-50 years old randomized to treatment, a female participant 45-50 years old is randomized to the placebo arm.
    - We generally refer to this as "exchangeability of treatment assignment in expectation" when we have sufficiently large numbers of people

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These arguments hold under perfect compliance

- In other words, people are assigned to an RCT arm, so these arguments are based on treatment assignment not necessarily on receipt/adherence to the actual treatment itself!
  - And IF there is perfect compliance to assigned treatment (intervention or placebo arm) all these arguments can then apply to the causal effect *of treatment* itself.
- How are the following achieved in an RCT via randomization:
  - Temporality
  - Consistency
  - Positivity
  - **Exchangeability**

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Example: Women With Hysterectomy (WHI); *JAMA* 2004

- **Context:** Despite decades of use and considerable research, the role of estrogen alone in preventing chronic diseases in postmenopausal women remains uncertain.
- **Objective:** To assess the effects on major disease incidence rates of the most commonly used postmenopausal hormone therapy in the United States.
- **Design, Setting, and Participants:** A randomized, double-blind, placebo-controlled disease prevention trial (the estrogen-alone component of the Women's Health Initiative [WHI]) conducted in 40 US clinical centers beginning in 1993. Enrolled were 10 739 postmenopausal women, aged 50-79 years, with prior hysterectomy, including 23% of minority race/ethnicity.
- **Intervention:** Women were randomly assigned to receive either 0.625 mg/d of conjugated equine estrogen (CEE) or placebo.
- **Main Outcome Measures:** The primary outcome was coronary heart disease (CHD) incidence (nonfatal myocardial infarction or CHD death). Invasive breast cancer incidence was the primary safety outcome. A global index of risks and benefits, including these primary outcomes plus stroke, pulmonary embolism (PE), colorectal cancer, hip fracture, and deaths from other causes, was used for summarizing overall effects.

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Example: Women With Hysterectomy (WHI); *JAMA* 2004

- What is the treatment/exposure?
- What is the outcome?

- Questions relevant to causal identification:
  - **Temporality**: does the treatment precede outcome?
  - **Consistency**: are people exposed to treatment in the same way?
  - **Positivity**: did everyone in the trial have a nonzero probability of receiving either treatment?
  - **Exchangeability**: are risk factors for the outcome balanced across arms of the trial?

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Assessing exchangeability in this trial

Characteristics	CEE (n = 5310)	Placebo (n = 5429)
Age at screening, mean (SD), y	63.6 (7.3)	63.6 (7.3)
Age group at screening, y, No. (%)		
50-59	1637 (30.8)	1673 (30.8)
60-69	2387 (45.0)	2465 (45.4)
70-79	1286 (24.2)	1291 (23.8)
Race/ethnicity, No. (%)		
White	4007 (75.5)	4075 (75.1)
Black	782 (14.7)	835 (15.4)
Hispanic	322 (6.1)	333 (6.1)
American Indian	41 (0.8)	34 (0.6)
Asian/Pacific Islander	86 (1.6)	78 (1.4)
Unknown	72 (1.4)	74 (1.4)
Hormone use, No. (%)		
Never	2769 (52.2)	2770 (51.1)
Past	1871 (35.2)	1948 (35.9)
Current†	669 (12.6)	708 (13.0)
Duration of prior hormone use, y, No. (%)‡		
<5	1352 (53.2)	1412 (53.1)
5-<10	469 (18.5)	515 (19.4)
≥10	720 (28.3)	732 (27.5)
Body mass index, mean (SD)§	30.1 (6.1)	30.1 (6.2)
Body mass index, No. (%)		
<25	1110 (21.0)	1096 (20.3)
25-29	1795 (34.0)	1912 (35.5)
≥30	2376 (45.0)	2383 (44.2)
Systolic BP, mean (SD), mm Hg	130.4 (17.5)	130.2 (17.6)
Diastolic BP, mean (SD), mm Hg	76.5 (9.2)	76.5 (9.4)
Smoking, No. (%)		
Never	2723 (51.9)	2705 (50.4)
Past	1986 (37.8)	2089 (38.9)
Current	542 (10.3)	571 (10.6)
Parity, No. (%)		
Never pregnant/no term pregnancy	489 (9.3)	461 (8.5)
≥1 Term pregnancy	4779 (90.7)	4932 (91.5)
Age at first birth, y, No. (%)‖		
<20	1193 (28.1)	1234 (28.0)
20-29	2846 (67.0)	2914 (66.1)
≥30	210 (4.9)	260 (5.9)

Characteristics	CEE (n = 5310)	Placebo (n = 5429)
Age at hysterectomy, y, No. (%)		
<40	2100 (39.8)	2149 (39.8)
40-49	2281 (43.2)	2275 (42.2)
50-54	501 (9.5)	566 (10.5)
≥55	401 (7.6)	404 (7.5)
Bilateral oophorectomy, No. (%)	1938 (39.5)	2111 (42.0)
Medical treatment, No. (%)		
Treated for diabetes	410 (7.7)	411 (7.6)
Treated for hypertension or BP ≥140/90 mm Hg	2386 (48.0)	2387 (47.4)
Elevated cholesterol levels requiring medication	694 (14.5)	766 (15.9)
Statin use at baseline	394 (7.4)	427 (7.9)
Aspirin use (≥80 mg/d) at baseline	1030 (19.4)	1069 (19.7)
Medical history, No. (%)		
Myocardial infarction	165 (3.1)	172 (3.2)
Angina	308 (5.8)	306 (5.7)
CABG/PTCA	120 (2.3)	114 (2.1)
Stroke	76 (1.4)	92 (1.7)
DVT or PE	87 (1.6)	84 (1.5)
Female relative had breast cancer, No. (%)	893 (18.0)	870 (17.1)
Fracture at age ≥55 y, No. (%)	676 (14.0)	643 (13.2)
No. of falls in last 12 mo, No. (%)		
0	3300 (67.0)	3230 (64.8)
1	975 (19.8)	1024 (20.5)
2	422 (8.6)	478 (9.6)
≥3	231 (4.7)	255 (5.1)

Abbreviations: BP, blood pressure; CABG/PTCA, coronary artery bypass graft/percutaneous transluminal coronary angioplasty; CEE, conjugated equine estrogen; DVT, deep vein thrombosis; PE, pulmonary embolism.
*Subgroup totals may not sum to number randomized because of missing data.

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Exchangeability in RCTs

- Exchangeability across arms in an RCT suggests that *aside from an effect of the treatment, the average risk of the outcome among the treated is the same as the average risk of the outcome among the untreated*
  - Bc, on average (aka in expectation) randomization balances covariates (confounders) between arms, even for unmeasured or unknown confounders
- Why might the pre-treatment risk in one group be different than in another?
  - For example, what if other causes of the outcome are more prevalent in one group than another…. How would this influence our ANALYTIC APPROACHES of ITT vs. Compliance-corrected methods?

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Using DAGs to understand this

- Consider a multi-site trial to test use of a new drug to reduce mortality risk among people who have been newly diagnosed with cancer
  - **Study Sample eligibility:** 45-65yo with incident cancer diagnosis, n=500
  - **Exposure:** Once/day New Drug X medication (vs. standard of care)
  - **Primary Endpoint:** One-year risk of mortality

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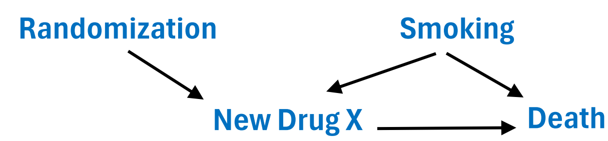
Using DAGs to understand this (cont.)

- What if this happens:



- In this version of events, those randomly assigned to NDX may or may not actually take it – and actually receiving/taking NDX might be influenced by behaviors (smoking) that also influence death.
  - For example, if more people in the treatment arm smoked compared to the placebo, then the risk of death in the treatment arm would be higher
  - Exchangeability matters because an ITT analysis examines the effect of randomization, whereas compliance-corrected analysis examines the effect of Injection!

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Intent to Treat Analysis in RCTs

- The ITT approach considers the effect of the intervention as it was "intended", <u>*not* as it actually happened.</u>
- Persons who were randomized to treatment and to control are analyzed as such even if they do not remain in those arms
  - This minimizes non-comparability between groups (since group comparability – exchangeability – was based on <u>original</u> *randomization* not on who did eventually take treatment vs. not)

	Dx	No Dx	LTFU	Total N	$PT_{\sim LTFU}$	PT_{LTFU}	PT_{Total}
Treatment	15	30	5	50	30	20	50
Placebo	20	20	10	50	25	25	50

ITT based findings:
CIR = 0.75
IDR = 0.75

Not employing ITT:
CIR = 0.67
IDR = 0.63

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Compliance-corrected analysis

- Randomization is broken in compliance-corrected analysis.
  - Remember: random assignment affects who gets the actual intervention
  - BUT other factors (smoking, age, other comorbidities) can affect compliance with randomization and therefore who actually receive (take/adhere to) the intervention
    - these other factors are not randomized between those who comply vs. don't comply to the intervention
- What implications does compliance correction have for consistency, positivity, exchangeability?
  - Can we assume that consistency is still met by design, in a compliance-corrected analysis?

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Problems & limitations in RCTs

- Non-compliance (and as-treated / compliance-corrected analyses)
- Loss to follow-up
- Dealing with non-compliance and loss-to-follow-up
- Adverse event monitoring and stopping
- When to not do a trial?

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Non-compliance

- Participants don't accept or adhere to an intervention (or standard of care or placebo) as they agreed to…
- The intention-to-treat analysis asks only about <u>treatment assignment</u>, not actual treatment.
  - Regardless of compliance to assigned treatment
  - Thus, non-compliance is typically ignored in this type of analysis
  - If we want to know the effect of actual treatment, irrespective of randomization, then we need to treat that analysis as an observational study and account for confounding

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Non-compliance (cont.)

- Assume that after randomization, participants don't adhere to their assigned treatment:
  - If we want to know the effect of treatment assignment → non-compliance would be treated as part of the total effect of treatment assignment. (i.e. daily oral PrEP regimen, but not everyone adheres to daily regimen)
  - If we want to know the effect of actual treatment and use the effect of treatment assignment **as a proxy** for the *effect of actual treatment* →then as non-compliance increases, treatment assignment will be a poorer estimate the effect of actual treatment

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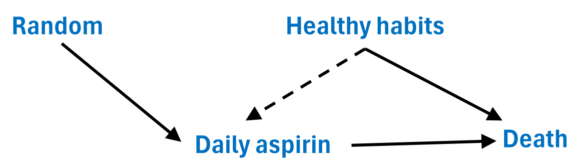
Confounding

- In RCTs – ITT analysis of treatment assignment is meant to reduce likelihood of confounded results
- Confounding can occur when the exposure is no longer treatment assignment (which is un-confounded) but actual treatment (which may well be confounded)



- That is, the effect of randomization (randomization to Daily Aspirin vs placebo) is not confounded by Healthy Habits. The effect of ***actually taking*** daily Aspirin is confounded, however!!

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Loss to follow-up (LTFU)

- Related conceptually to non-compliance
- Participants will drop out of trial while it is in progress
  - Outcome data no longer available → missing data
- LTFU may lead to bias if there is a relationship between incidence of outcome and LTFU
  - E.g., people who are about to die more likely to be LTFU (or do not return to clinic because they are dead)
  - This becomes a selection bias
- What are the implications of LTFU?
  - Reduces study power!!

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LTFU (cont.)

- When is leaving a study considered LTFU?
  - An applied definition of LTFU: *Follow-up visits will be scheduled for every 6 months from a fist (baseline) visit. Participant may be seen +/- 30 days from 6 month date. If visit is missed, then follow-up with 3 phone calls and a home visit over an additional 6 month period. If participant is not located after those attempts, then they are considered LTFU*
  - So, when is participant considered LTFU?
    - At 6 months? Or at 6months + 6 months = 12 months?
    - Why?

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Handling non-compliance and LTFU

- Consider inclusion/exclusion criteria
- Clearly lay out trial duration during screening and consent
- Collect as much data as possible (feasible) at baseline
  - Especially data that might help understand LTFU or non-compliance
- Try to prevent LTFU**
- Try to assess why people are dropping out (when possible)
- Assess intervention compliance constantly!!
    - Survey
    - Bio samples
    - MEMSCap system automatically records when a pill bottle is opened (but does that correspond to medication taken?)

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Relatedly: right-censoring

- Trials are established with a set follow-up duration
- At the end of follow-up period, not everyone has had the event
  - These people are typically regarded as "censored": we know their event time is more than 1 year, but we don't know when the event occurs
  - This is called "administrative right censoring," and is usually considered to be uninformative
- The other kind of right-censoring is when someone disappears from the trial – for reasons non-administrative
  - Unlike administrative right censoring, cannot be presumed uninformative (if you have good data that can help understand reasons for drop out/LTFU)

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Biases in RCTs (well, really in all epi studies…)

Recall, these are:

- Missing data
- Confounding
- Selection bias
- Measurement error
- Non-differential misclassification

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Missing data in trials

- In RCTs
  - Intervention treatment assignment is under the investigator's control
  - So "exposure" status is "known" (compliance?)
  - Outcome information may be missing
    - By chance – lost/damaged samples
    - LTFU!
      - Reminder: when does LTFU become a bias?

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Remember: when does LTFU lead to selection bias?

- LTFU will generally cause bias if people become lost <u>because they experienced the outcome</u>
    - Caveat: If losses are linked to treatment assignment only (side-effects in active treatment arm → LTFU) then there might not be bias.

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Measurement error

- If you measure variables with error in your trial (for example: treatment assignment, outcome, time of outcome...) then your analyses which rely on those variables will be wrong!
- Closely related: recall, if you want to interpret the ITT as an estimate of the true effect of the treatment (and not just of treatment assignment) then non-compliance becomes a problem.
  - For example, if persons assigned to active drug do a poor job taking it (take only 50% of doses); or worse, take none of it, then "assigned to active drug" doesn't have that much to do with the effect of the drug itself.
  - This can be viewed as a problem of measurement error. You can also think of this in terms of consistency: is this a relevant variation in how the treatment was received? (Yes!)

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Non-compliance & measurement error

- So, non-compliance in a trial (in an ITT analysis) means that some people assigned to treatment, and analyzed as treated, are actually untreated. And vice-versa.
  - **What does this mean if you want to interpret the results of that ITT in terms of the effect of the actual treatment?**
  - Well, mixing your groups has the effect of making the assigned-treated group more similar to the assigned-untreated group. If the two groups are more similar, measures of contrast (e.g., risk differences) are going to be:

    (a) Closer to the null?

    (b) Further from the null?

    (c) Generally unaffected? (What if the results are null?)

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Non-differential misclassification

- If non-compliance is independent of outcome risk, then this can be considered a form of "non-differential misclassification"
  - Misclassification is not differential by outcomes status.
- Non-differential misclassification in two categories *generally* leads to bias toward the null
  - because it makes the two categories more similar to each other.
  - In this case, some of the people analyzed as group 1 actually experienced the effects of being in group 2.
- In more than two categories, the direction of bias is less easily predicted.
  - It depends on which category is used as the reference category (denominator).

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ITT vs. compliance corrected vs. per-protocol analyses

- **ITT analyses** = participants analyzed based on the arm to which they were randomized
- **Compliance corrected** = (aka "as treated") analysis involves analyzing RCT participants according to the treatment that they took
- **Per-protocol analyses** = (aka "on treatment") includes only participants who strictly adhered to the study protocol, often excluding non-compliant subjects
  - Typically, only includes a subset of trial participants referred to as the per protocol population
  - These contrasts will be affected by selection bias!

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Comparing the 3 types of analyses

ITT

- **Inclusion:** by randomization status
- **Bias Risk:** minimizes bias by increasing comparability between arms
- **Purpose:** evaluate efficacy, broadly speaking

Per-protocol

- **Inclusion:** excludes non-compliers
- **Bias:** increased bc non-compliers dropped, therefore comparability of arms reduced
- **Purpose:** efficacy, under ideal conditions

Compliance-corrected

- **Inclusion:** reclassifies participants based on actual intervention compliance
- **Bias:** biased if compliance associated with outcome risk
- **Purpose:** efficacy, under ideal conditions

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An example...

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RESEARCH HIGHLIGHT | 20 February 2026

AI stethoscope trial highlights the perils of implementation gaps

When used as intended, an AI stethoscope enhanced the detection of cardiovascular disease – but low uptake and workflow challenges hampered effectiveness in a large pragmatic trial.

By Karen O'Leary

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AI-enabled stethoscopes show promise for improving diagnosis of cardiovascular conditions

by Emily Medcalf

29 January 2026

Triple cardiovascular disease detection with an artificial intelligence-enabled stethoscope (TRICORDER) in the UK: a cluster-randomised controlled implementation trial

Mihir A Kelshiker, Patrik Bächtiger*, Camille F Petri, Saloni Nakhare, Josephine Mansell, Karanjot Chhatwal, Abdullah Alrumayh, Jahed Zaman, Moulesh Shah, Holly Young, Helena Roy, Melanie T Almonte, Céire Costelloe, Yasmin Razak, Azeem Majeed, James P Howard, Carys Barton, Daniel B Kramer, Carla M Plymen, Nicholas S Peters*

Kelshiker, Mihir A et al. Triple cardiovascular disease detection with an artificial intelligence-enabled stethoscope (TRICORDER) in the UK: a cluster-randomised controlled implementation trial. The Lancet, Volume 407, Issue 10529, 704 - 715

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Summary

Background Early detection of cardiovascular disease is a global public health priority. Artificial intelligence (AI)-enabled stethoscopes offer robust performance characteristics in point-of-care detection of heart failure, atrial fibrillation, and valvular heart disease (VHD). We conducted a pragmatic, cluster-randomised controlled implementation trial to determine the real-world effect and implementation challenges of AI-stethoscopes.

Methods UK primary care practices were cluster randomised 1:1 to intervention (training and implementation in use of AI-stethoscopes in routine care) or control (routine care). Given the nature of the intervention, masking of participants (practices, clinicians, and patients) was not feasible. During cardiac examinations, the AI stethoscope recorded 15 s of single-lead electrocardiogram and phonocardiogram signals for input to three AI algorithms that returned binary predictions for the presence or absence of reduced left ventricular ejection fraction (≤40%), atrial fibrillation, and VHD (all with regulatory approval). The primary endpoint was incidence of any newly coded diagnosis of heart failure (all subtypes), expressed per 1000 patient-years (incidence rate ratio [IRR]), derived from a UK National Health Service Secure Data Environment. A coprimary endpoint stratified detection of heart failure by place of diagnosis (community-based *vs* via hospital admission). Secondary endpoints included atrial fibrillation and VHD detection rates, performance characteristics of the AI-stethoscope, use rates, and clinician-reported implementation barriers and enablers.

Findings Between Oct 30, 2023, and May 22, 2024, 205 practices were randomly assigned (96 to the intervention arm [701 933 registered patients] and 109 to the control arm [851 242 registered patients]). Intervention practices recorded 12 725 patient examinations with the AI-stethoscope, across 972 clinical users. Intention-to-treat analysis found heart failure detection did not differ between groups (IRR 0·94 [95% CI 0·86–1·02]); with no difference in community-based or hospital-based diagnoses (p>0·05).

Interpretation Implementation of an AI stethoscope in routine primary care did not significantly increase detection of heart failure or increase community-based diagnosis after 12 months of implementation. AI stethoscope use was independently associated with significantly higher detection rates of heart failure, as well as atrial fibrillation and VHD. This randomised controlled implementation trial establishes a pragmatic design with randomisation that generates real-world data essential for understanding and overcoming the barriers to implementation of innovation in health care.

Funding National Institute for Health and Care Research, British Heart Foundation, and Imperial Health Charity.

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Where to begin....

- Communicating results:
  - Why aren't the ITT analysis findings reported in these headlines?
    - *"Implementation of an AI stethoscope in routine primary care did not significantly increase detection of heart failure or increase community-based diagnosis."*
- This is an example of how per-protocol analyses can be challenging
  - Provider decision to use an AI stethoscope is not random…how are pt and provider factors that drive decision to use an AI stethoscope controlled for in these comparisons?
    - Analysis only controlled for age, sex, ethnicity, certain comorbidities, and socio-economic position..but what about residual confounding?
    - Factors that are associated with outcomes more likely in per-protocol groups
    - Providers were selectively using AI stethoscope on dramatically older, sicker patients and more likely to find heart problems in these patients because they were more likely to develop them!!!

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Randomized Trials – Part II
Additional information and types of trials

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Additional Useful Terms

- **Parallel Design:**
  - A parallel design in reference to a clinical study where two or more groups of participants receive different interventions.
  - Participants are assigned to one trial arm at the beginning of the trial and continue in that arm throughout the length of the trial.
  - Group assignment is randomized.
  - Study participants are only exposed to the treatment that is assigned to the particular study arm they are enrolled in.
    - Example: A two-arm parallel design clinical trial involves two groups of participants. One group receives drug A, and the other group receives drug B. So during the trial, participants in one group receive drug A "in parallel" to participants in the other group, who receive drug B.

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Additional Useful terms (cont.)

- **Run-in-period or washout period:** A pre-defined period before the actual start of the intervention where no intervention (typically a drug) is given to screen out for ineligibility/noncompliance
  - Particularly relevant for pharmaceutical trials of medications to assess questions such as: *How is the drug metabolized? Does the drug have a measurable effect on the disease it's designed to treat? Is that measurable effect better than the already available treatments? Are the side effects manageable?*
  - Want to be sure the effects we measure are caused by the investigational drug, not the latent effects of some medication taken before the start of the trial.
  - Also want to protect people from potential drug interactions (known or unknown).
  - Disadvantages: length of washout period and requirement to engage in a wash-out periods may deter study enrollment.

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Additional Useful terms (cont.)

- **Contamination**: Situation where the experimental (intervention) group influences the control group or vice verse and there is experimental "contamination"
  - Transparency in trials is increasingly important to demonstrate research integrity.
  - But: Transparency has the potential to create contamination risks
  - Examples:
    - When social interactions between individuals randomly assigned to different treatment conditions causes some individuals to receive features of a treatment to which they were not assigned.
    - Situations in which health professionals are delivering both intervention and standard of care treatments and there is communication among clinicians and participants from the different trial arms. *(Magill N, et al. A scoping review of the problems and solutions associated with contamination in trials of complex interventions in mental health. BMC Med Res Methodol **19**, 4 (2019)*

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Useful terms (cont.)

	Efficacy study	Effectiveness study
Question	Does the intervention work under ideal circumstance?	Does the intervention work in real-world practice?
Setting	Resource-intensive 'ideal setting'	Real-world everyday clinical setting
Study population	Highly selected, homogenous population Several exclusion criteria	Heterogeneous population Few to no exclusion criteria
Providers	Highly experienced and trained	Representative usual providers
Intervention	Strictly enforced and standardized No concurrent interventions	Applied with flexibility Concurrent interventions and cross-over permitted

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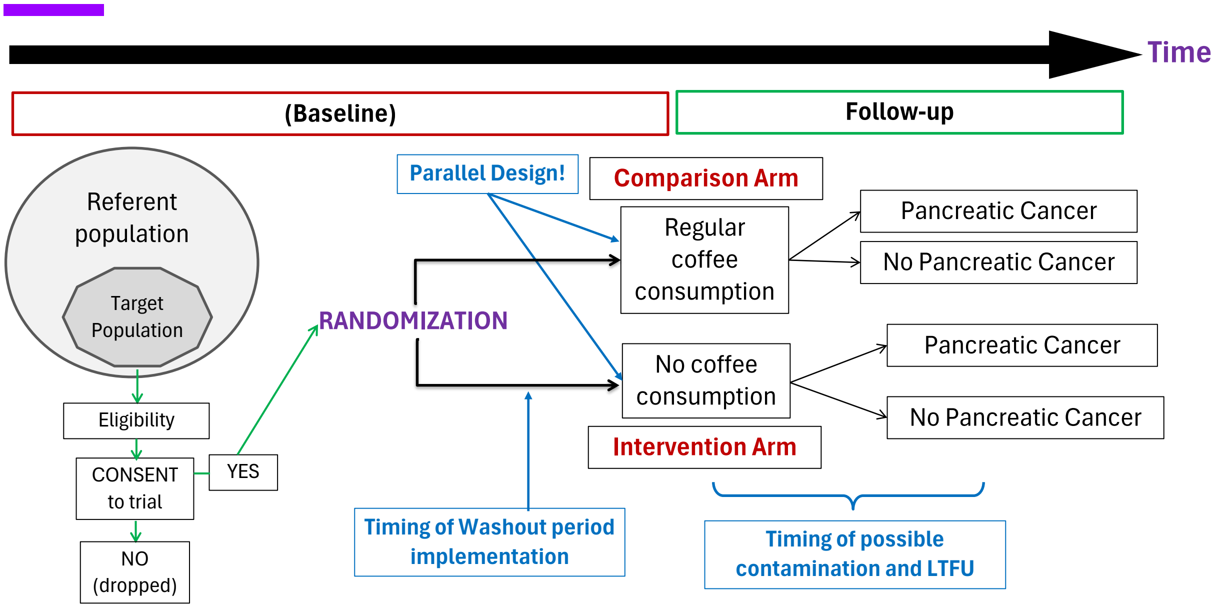
RCT schematic: these terms in action!



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Superiority vs. Non-Inferiority RCTs

- **Superiority Design: (traditional RCT)**
  - Designed to provide evidence that a new treatment is more efficacious than the control (maybe a placebo) or standard of care.
- **Equivalence/Non-inferiority:**
  - Provide evidence that a new treatment (i.e. an easier or cheaper tx) <u>is not worse</u> than the standard by more than a pre-established margin (known as the margin of equivalence or non-inferiority), in other words, is the new tx "just as good" as the current tx
  - Can't statistically prove equivalency -- only show that difference is less than something with specified probability
    - **Issues to deal with:**
    - Small sample size, leading to low power and subsequently lack of significant difference, does not imply "equivalence"
    - Treatment margin can be arbitrary

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Cluster Randomized Trials

- Similar in concept to individual level studies, but here units of assignment may be schools, worksites, clinics, or whole communities, and the units of observation are the students, employees, patients, or residents within those groups.
  - Groups may only be randomized to <u>one intervention condition</u>, so members <u>within groups are only randomized to the same one intervention condition</u>
    - But note: members within a given group may be more like each other and different from members of a different group. Therefore, there may be less variation by member within groups than across group.
      - Can employ a *priori* matching, a *priori* stratification, or constrained randomization to balance potential confounders, to reflect the hierarchical structure of the design in the analytic plan

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Cluster Randomization Designs

- Groups (clinics, communities) are randomized to treatment or control
  - Examples:
    - Community trials on fluoridization of water
    - Breast self examination programs in different clinic setting in USSR
    - Smoking cessation intervention trial in different school district in Washington state
  - Advantages:
    - Sometimes logistically more feasible
    - Avoid contamination
    - Allow mass intervention, thus “public health trial”
  - Disadvantages:
    - Effective sample size less than number of subjects
    - Many units must participate to overcome unit-to-unit variation, requiring larger sample size
    - Need cluster sampling methods

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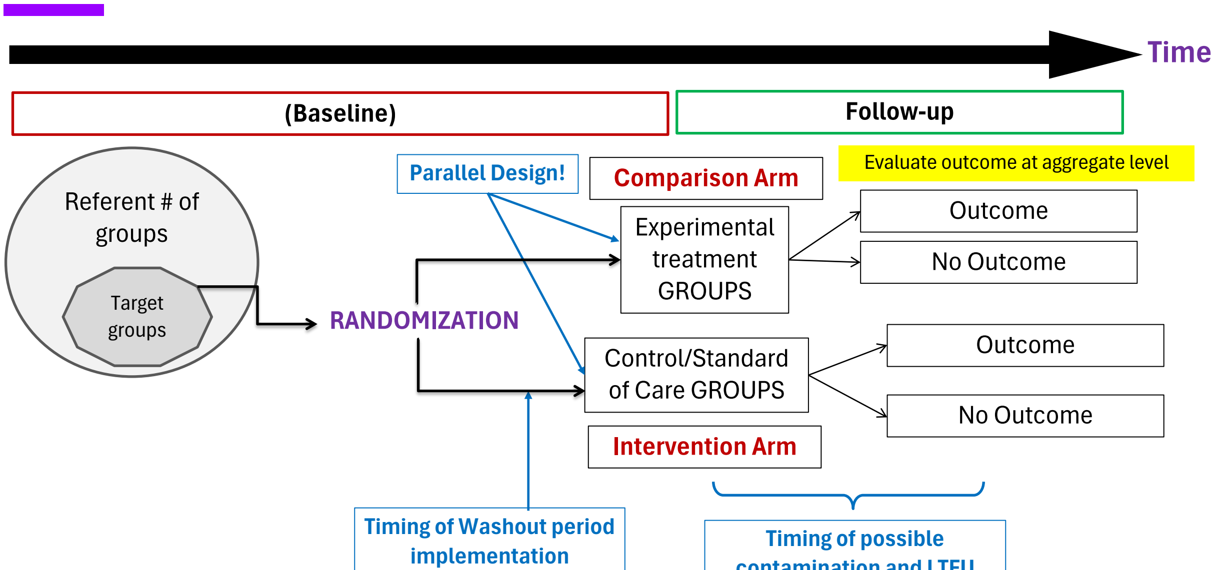
Cluster RCT schematic



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Crossover design characteristics

- **Overall Approach**:
  - Sequential treatments for *each* participant
  - Patients act as their own controls
- **Advantages:**
  - each participants serve as own controls and error variance is reduced, thus reducing sample size needed bc paired statistical analysis markedly increases power
  - each participants receive treatment (at least some of the time)
  - statistical tests assuming randomisation can be used
  - blinding can be maintained
  - good when treatment effect small *versus* population variability

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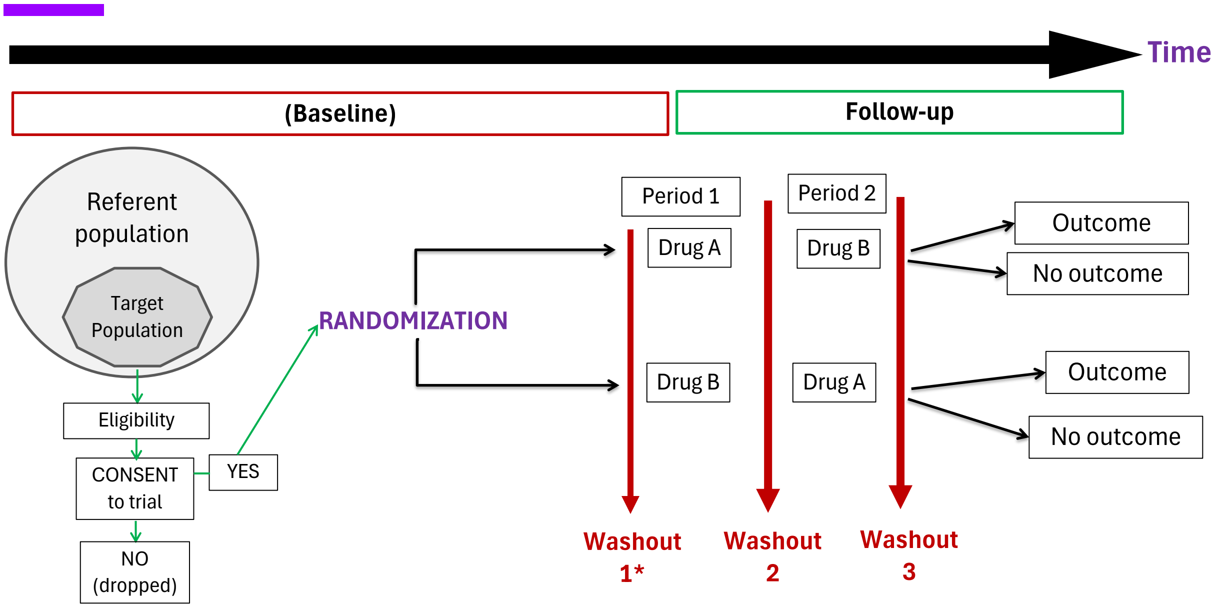
RCT schematic: Crossover designs



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Crossover design characteristics *(cont.)*

- Disadvantages:
  - Assumes underlying disease state is static
  - Assumes lack of carry-over effect of initial treatment
  - May require a treatment-free washout period that can be lengthy
  - Evaluate markers rather than "hard" outcomes
  - Cannot be used for one-time treatments such as surgery
  - Cannot be used for treatments with permanent effects
  - All participants must receive the alternative treatment but is there also a placebo?

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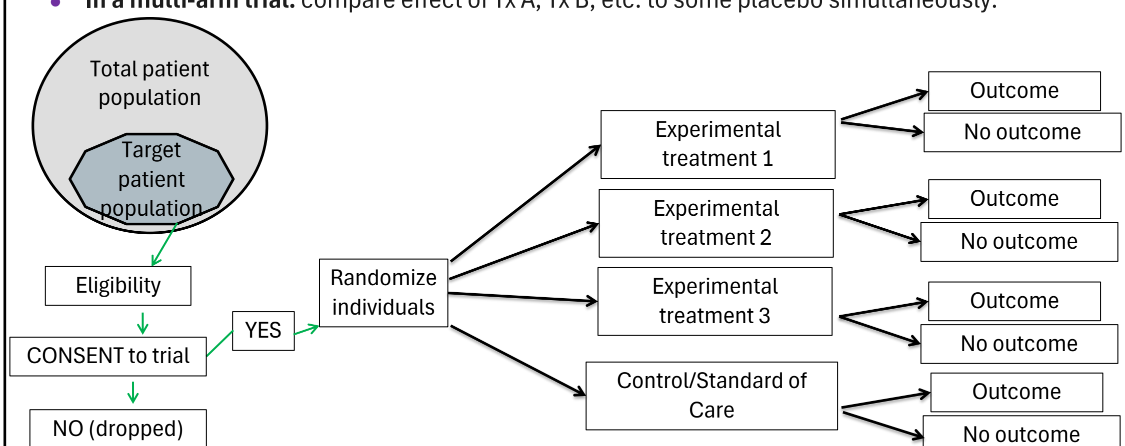
RCT schematic: Multi-arm trials

- When several intervention or treatment options are available, different design options are available.
- **In a multi-arm trial:** compare effect of Tx A, Tx B, etc. to some placebo simultaneously:



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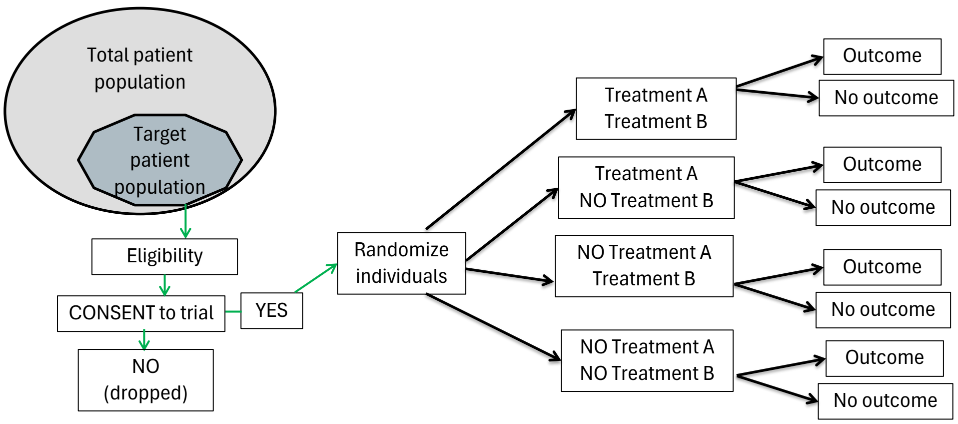
RCT schematic: Factorial Designs

- **Factorial designs:**
  - The most common is a 2x2 design
  - Patients will either take treatment A, B, both treatments or none



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Sample size considerations in RCTs

- RCTs employ comparisons of time to event distributions
  - Primary endpoint is the time to an event
  - Compare the survival distributions
  - Measure of treatment effect is the ratio of the hazard rates
  - Must also consider the length of follow-up AND loss to follow-up
- Therefore, sample size for a RCT requires knowledge of:
  - Expected number of events
  - Expected timing of events (i.e. over the duration of the follow-up period) – is the hazard rate (time to events) static or dynamic over the follow-up period?
  - Potential non-compliance – will the drop out rate vary over the follow-up period?
- Ethically, sample size must be large enough to achieve stated goals with reasonable probability (power)
- Sample size estimates are only approximate due to uncertainty in assumptions listed above
- Need to be conservative but realistic!!

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Background

- Quasi-experimental designs can be used when classic RCT design is not feasible or ethical
  - Therefore, need to minimize, as possible, potential risks to internal validity arising from selection bias and confounding
- Key Characteristics of quasi-experimental studies:
  - **No Randomization:** Participants not randomly assigned to experimental/comparison groups

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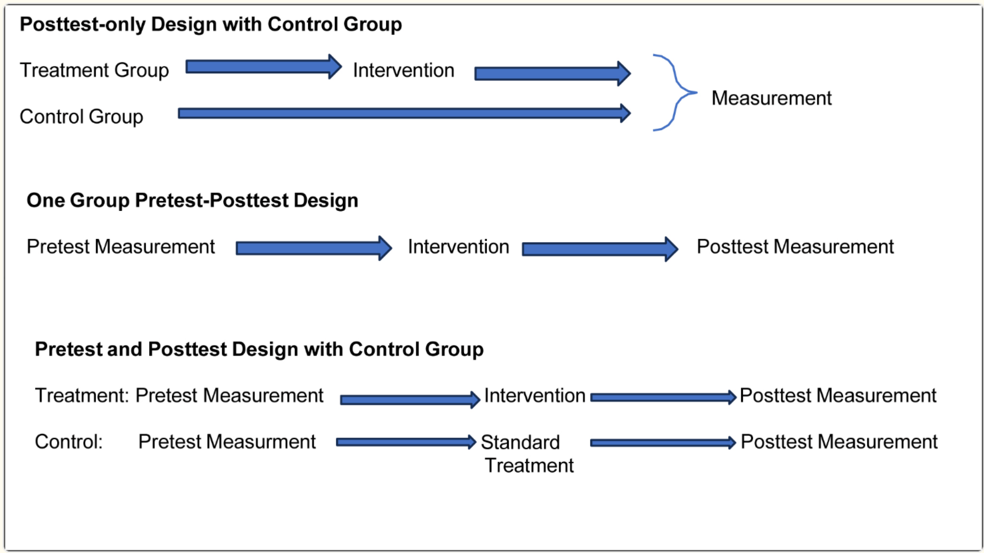
Non-equivalent control groups designs



Capili B, Anastasi JK. An Introduction to Types of Quasi-Experimental Designs. Am J Nurs. 2024 Nov 1;124(11):50-52.

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Regression Discontinuity Design (RDD)

- A type of pretest–posttest design that attempts to determine causal effects of an intervention by assigning a cutoff or threshold above or below which an intervention is assigned (or occurs)
  - comparing observations lying close to either side of the threshold to estimate the average treatment effect in environments when randomization is not possible
- One key assumption of RDD: treatment assignment is "as good as random" at the threshold for treatment
  - If this assumption holds, then those who just barely received treatment are comparable to those who just barely did not receive treatment
    - Example: Getting a scholarship requires a GPA of 3.5. And since GPAs are a function of course types, grading rubrics, and student performance, then they are somewhat random.

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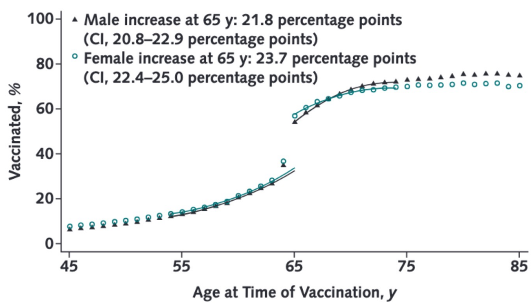
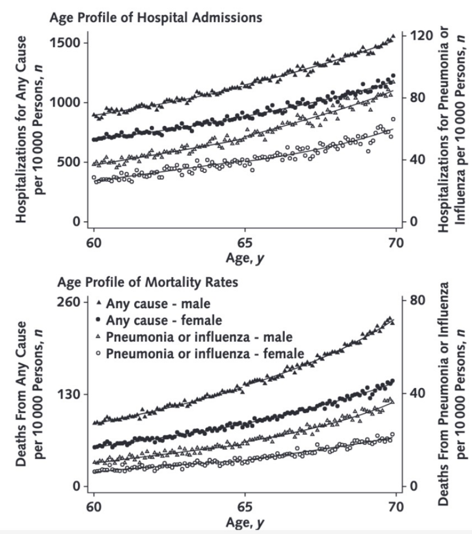
Flu vaccination and hospitalization and mortality: RDD analysis

Figure 1. Age profile of vaccination status.



- What could explain these finding?

Figure 2. Age profile of hospital admissions (*top*) and mortality rates (*bottom*).



Anderson et al. Ann Intern Med. 2020 Apr 7;172(7):445-452.

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Difference in Differences (DiD)

- DiD uses longitudinal data (cohort, panel or repeat cross-sectional) from treatment and control groups to obtain an appropriate counterfactual to estimate a causal effect.
- Goal is to isolate the causal effect of an intervention (e.g., policy, law, or program) by controlling for factors that change over time
  - compare changes in outcomes over time between a population that is enrolled in a program (the intervention group) and a population that is not (the control group)

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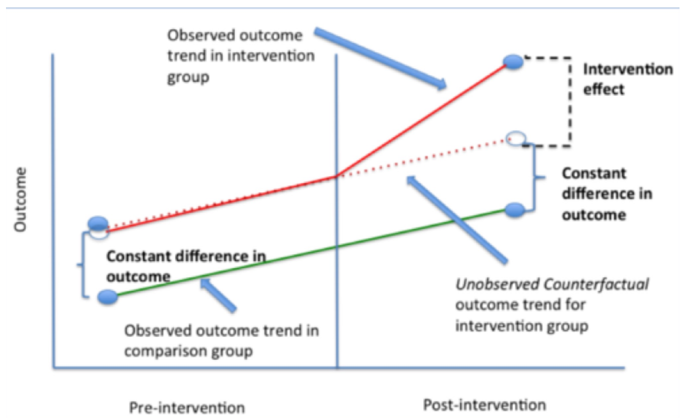
Key DiD Assumptions

- Intervention unrelated to outcome at baseline (allocation of intervention was not determined by outcome)
- SUTVA (Stable-Unit-Treatment-Value-Assumption)
- **Parallel Trends - requires that in the absence of treatment, the difference between the 'treatment' and 'control' group is constant over time



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Effect of income-supplementation on food insecurity: A DiD analysis

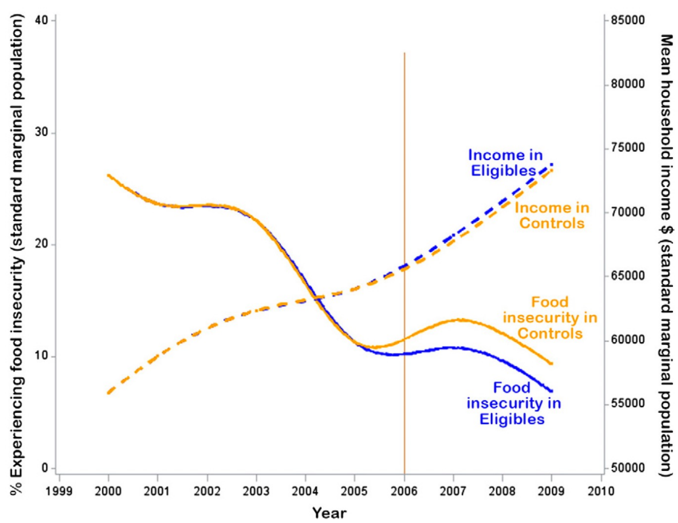

Fig. 3. Outcome trends in eligible versus controls standardized to the covariate distribution of the full population (predicted for the mean standard population).

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